Student Details

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- **Course/Program:** [Data Analytics]
- **Internship Organization:** [Unified Mentor]
- **Internship Duration:** [2 January–2 April]
- **Project Title:** SUPPLY CHAIN USING POWERBI DASHBOARD

1. Introduction

In modern organizations, supply chain operations generate a massive amount of data every day. Activities such as procurement, production, inventory management, distribution, and delivery continuously create operational information that can be analyzed to improve business performance.

However, raw supply chain data is often complex and difficult to interpret without proper analytical tools. Business intelligence platforms allow organizations to transform raw data into meaningful insights that support strategic decision-making.

This project focuses on designing and developing a **Supply Chain Management Dashboard using Microsoft Power BI**. The dashboard provides visual insights into supply chain activities including inventory levels, supplier performance, transportation efficiency, order fulfillment, and operational costs.

By using interactive visualizations, the dashboard enables stakeholders to monitor performance indicators and identify areas for improvement in supply chain operations.

2. Project Objectives

The main objectives of this project are:

- To analyze supply chain data using business intelligence tools.
- To develop an interactive **Power BI dashboard** for supply chain monitoring.
- To evaluate supplier performance and transportation efficiency.
- To monitor product demand and inventory availability.
- To identify trends in revenue, costs, and order fulfillment.
- To support data-driven decision making for supply chain optimization.

3. Tools and Technologies Used

Tool	Purpose
Microsoft Power BI	Data visualization and dashboard creation

Microsoft Excel / CSV Dataset	Data storage and preprocessing
DAX (Data Analysis Expressions)	Calculated fields and measures
Power Query	Data cleaning and transformation

4. Dataset Description

The dataset used in this project represents the supply chain operations of a **Fashion and Beauty startup company** dealing with cosmetic products. The dataset contains multiple attributes related to product distribution, supplier management, and logistics.

Important Dataset Fields

Product Information

- Product Type
- SKU
- Price
- Availability
- Number of Products Sold
- Revenue Generated

Customer and Market Data

- Customer Demographics
- Location

Inventory Data

- Stock Levels
- Order Quantities
- Lead Time

Supplier Information

- Supplier Name
- Supplier Location
- Production Volume
- Manufacturing Lead Time
- Manufacturing Costs

Quality Control

- Inspection Results
- Defect Rates

Transportation and Logistics

- Shipping Carriers

- Shipping Costs
- Transportation Modes
- Delivery Routes
- Shipping Time

This dataset enables analysis across multiple dimensions of supply chain performance.

5. Data Preparation

Before building the dashboard, the dataset was cleaned and transformed using **Power BI Power Query**.

Data Preparation Steps

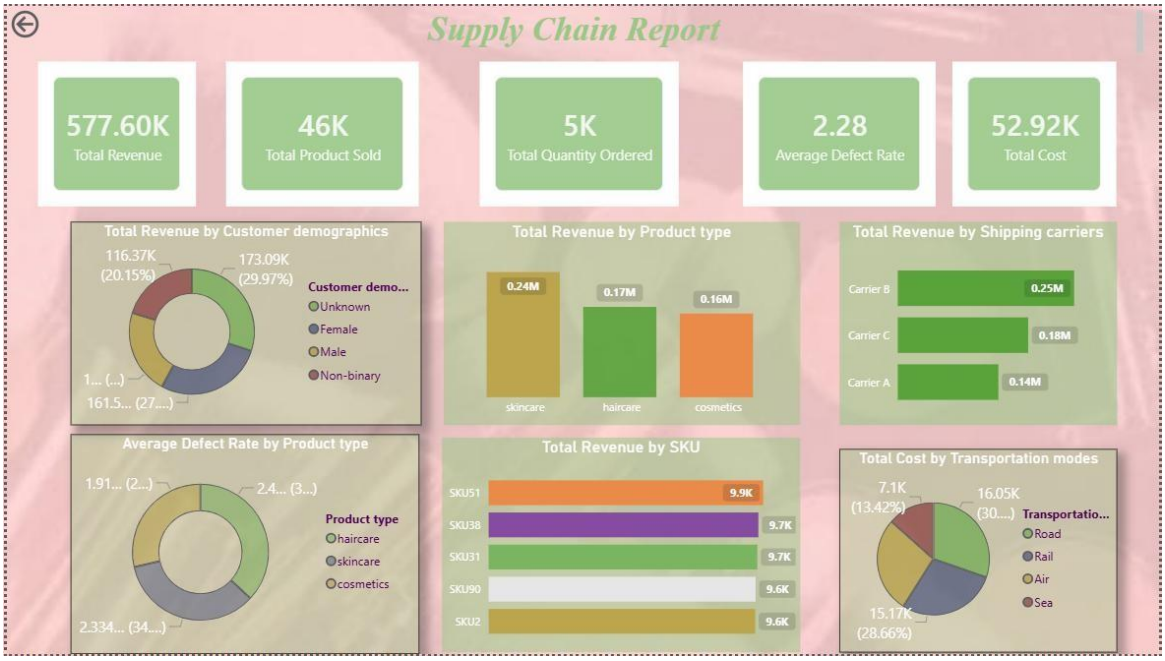
1. Import the dataset into **Power BI Desktop**.
2. Remove duplicate and incomplete records.
3. Format columns into appropriate data types.
4. Handle missing values where necessary.
5. Create relationships between related tables (if multiple datasets are used).
6. Generate calculated measures using **DAX**.

After cleaning and transforming the dataset, it becomes ready for visualization and analysis.

6. Power BI Dashboard Development

The dashboard was designed by creating multiple visualizations that represent key aspects of the supply chain.

The dashboard consists of the following analytical sections:



6.1 Inventory Management Analysis

Inventory management is critical for ensuring product availability while avoiding excessive stock holding.

Visualizations Used

- **Bar Chart**
  - Inventory levels by product type
  - Warehouse stock comparison
- **Line Chart** ○ Inventory trend over time

Key Insights

- Identify products with low stock availability
- Track stock fluctuations
- Detect inventory demand patterns

6.2 Order Fulfillment Analysis

This section analyzes the efficiency of order processing and delivery performance.

Visualizations Used

- **Line Chart**
  - Orders fulfilled over time
- **Bar Chart** ○ On-time delivery rate by product

Insights Generated

- Percentage of orders delivered on time

- Order volume trends
- Customer demand patterns

6.3 Supplier Performance Analysis

Supplier performance is evaluated based on delivery time, defect rate, and manufacturing costs.

Visualizations Used

- **Bar Chart** ○ Average delivery time by supplier
- **Line Chart** ○ Defect rates over time

Insights Generated

- Identify high-performing suppliers
- Detect suppliers with quality issues
- Monitor manufacturing efficiency

6.4 Transportation Efficiency

Transportation plays an important role in supply chain performance.

Visualizations Used

- **Scatter Plot** ○ Transit time vs shipping carriers
- **Line Chart**
    - Delivery performance over time

Insights Generated

- Compare performance of different shipping carriers
- Identify delays in product delivery
- Improve logistics planning

6.5 Supply Chain Cost Analysis

This section focuses on analyzing operational costs associated with the supply chain.

Visualizations Used

- **Pie Chart** ○ Distribution of supply chain costs
- **Line Chart** ○ Cost trends over time

DAX Measure

Total Supply Chain Cost

```
Total Cost = SUM(Costs[Amount])
```

Insights Generated

- Identify major cost drivers
- Track cost changes over time
- Optimize transportation and storage expenses

7. Dashboard Features

The Power BI dashboard includes several interactive features:

- **Dynamic Filters (Slicers)** ○ Product type ○ Supplier ○ Region ○ Date
- **Interactive Visualizations** ○ Drill-down capabilities ○ Cross filtering between charts
- **Real-Time Data Insights** ○ Dynamic updates based on selected filters

These features allow users to explore supply chain data in a flexible and intuitive manner.

8. Key Business Insights

After analyzing the supply chain dataset, several important insights were observed:

- Inventory levels remain stable for most product categories.
- Certain suppliers demonstrate faster delivery performance compared to others.
- Transportation costs represent a significant portion of operational expenses.
- Some routes show longer transit times, indicating potential logistics inefficiencies. ▯ Demand trends vary across product categories and customer demographics.

These insights help organizations improve operational efficiency and reduce costs.

9. Conclusion

The **Supply Chain Management Dashboard developed using Power BI** provides a comprehensive overview of supply chain operations. By integrating multiple data sources and visualizing key performance indicators, the dashboard enables businesses to monitor supply chain performance effectively.

The use of Power BI allows decision-makers to explore data interactively and gain actionable insights into inventory management, supplier efficiency, order fulfillment, and logistics performance.

Overall, this project demonstrates how business intelligence tools can transform raw supply chain data into meaningful analytics that support better operational decisions and strategic planning.